

In-term Test Q4 2025-2026

*This paper consists of **SIX** practice questions, carrying a total of 100 marks. Questions 1 and 2 are worth 10 marks each, while questions 3 to 6 carry 20 marks each.*

Students have 90 minutes to answer all the questions. No extra time is given for reading the questions. Additional Exam Booklet is provided on Moodle. Students use computers with MySQL or SQL Server to complete the questions.

Students must submit the Exam Answer via Moodle. (script .sql and report as .pdf)

File Name format: MIS443_ID_FullName.zip.

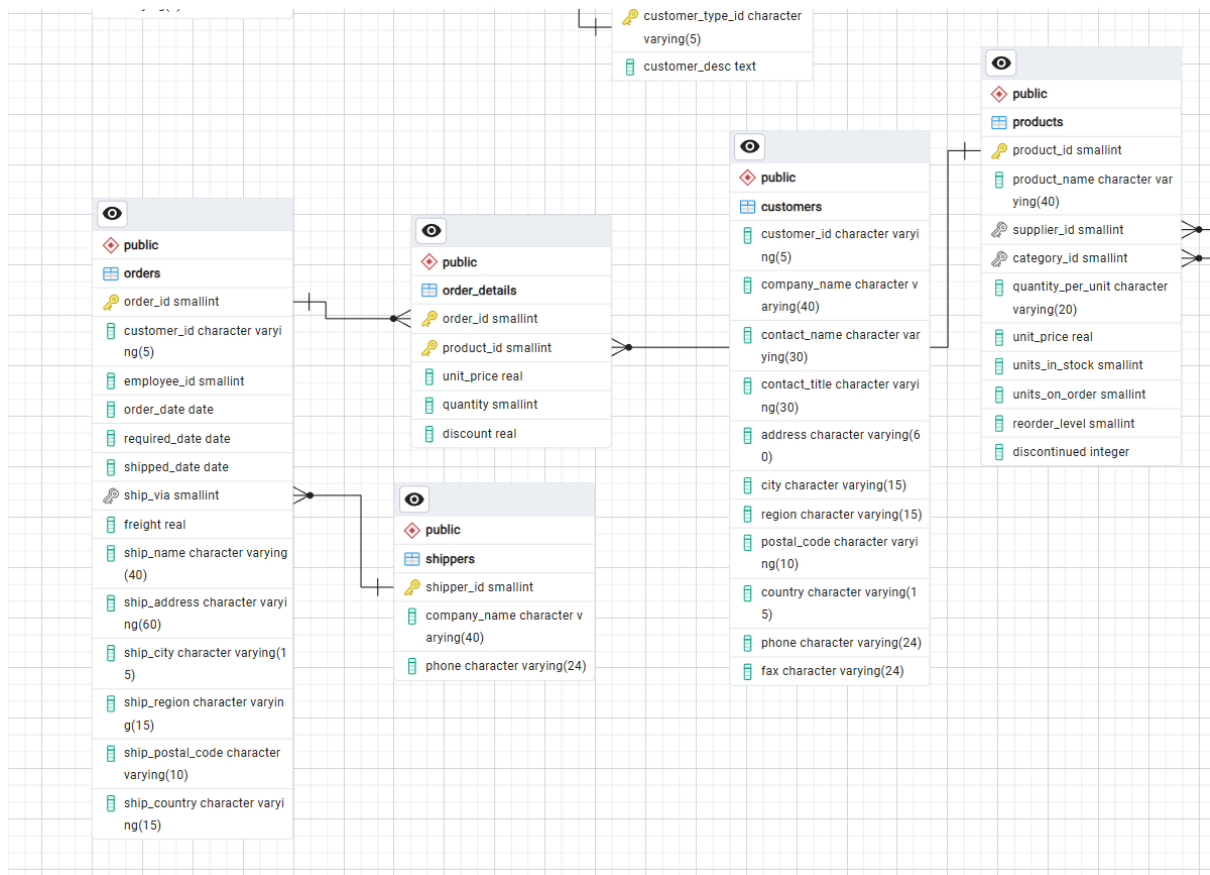
Scenario:

This dataset represents Northwind Traders, a mid-sized international trading company that sells food and beverage products to business customers. The database captures essential information about customers, orders, products, and shipping activities. The primary goal of this database is to support data-driven insights into customer purchasing behavior, order activity, and delivery operations, enabling the company to improve sales performance, optimize logistics, and make informed operational decisions.

- ✓ Customers: Stores each customer's unique ID and company information (e.g., company name, contact details, location). This table supports customer segmentation and relationship management.
- ✓ Orders: Records each order placed by a customer, including order date, shipping date, and shipping method. It provides the foundation for analyzing order frequency and delivery performance.
- ✓ Order_Details: Contains line-item details for each order (products, quantities, unit prices, and discounts). This table supports transaction-level analysis of product demand and sales patterns.
- ✓ Products: Lists all products sold by Northwind, including product name and related product attributes. It is used to analyze demand across different products.
- ✓ Shippers: Contains shipping partner information used to deliver orders. This table supports evaluation of logistics and shipping performance.

In addition, students must create a new table named "students". This table should include three attributes: studentid, fullname, and email. You may decide the appropriate data types and length for each attribute.

Note: The SQL script for loading the Northwind tables (include 14 tables) is available on Moodle. Only the "students" table must be created entirely by the student.



ER Model (five main tables)

Question 1 (10 marks): Database Setup (10 marks)

Using PostgreSQL and pgAdmin:

(a) Create a database named yourfullname. Then move all Northwind tables into this schema. (5 marks)

(b) Create a new table called students inside schema exam with the following columns:

Studentid: 5-digit number, Primary Key

Fullname: Required

Email: Must be unique

Insert your own information into the table. (5 marks).

-- Test results

SELECT table_name, column_name, data_type

FROM information_schema.columns

WHERE table_schema = 'public'

ORDER BY table_name, ordinal_position;

Sample Output:

Results 1: Verify result from load and create database

The screenshot shows the pgAdmin 4 interface. The SQL editor contains the following query:

```
/* (a) Create a database named yourfullname. Then load all Northwind tables into this schema. (5 marks)
Use file "Northwind.sql"
*/
select table_name, column_name, data_type
from information_schema.columns
where table_schema = 'public'
order by table_name, ordinal_position;
```

The query results are displayed in the Data Output pane, showing 92 rows. The results are as follows:

table_name	column_name	data_type
categories	category_id	smallint
categories	category_name	character varying
categories	description	text
categories	picture	bytea
customer_customer_de...	customer_id	character varying
customer_customer_de...	customer_type_id	character varying
customer_demographics	customer_type_id	character varying
customer_demographics	customer_desc	text
customers	customer_id	character varying
customers	company_name	character varying
customers	contact_name	character varying
customers	contact_title	character varying
customers	address	character varying
customers	city	character varying

Total rows: 92

Results 2: Verify result from table student

The screenshot shows the pgAdmin 4 interface. The SQL editor contains the following query:

```
-- (d) Verify the result
SELECT * FROM students;
```

The query results are displayed in the Data Output pane, showing 1 row. The results are as follows:

studentid	fullname	email
12345	VANVUQUYNHNU	nhu.vanvu.bbs22@ehu.edu...

Total rows: 1

Successfully run. Total query runtime: 1 secs 397 msec. 1 rows affected.

Question 2 (10 marks): Write an SQL query to find the top 5 customers who placed the highest number of orders.

Expected Output:

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following query:

```
-- Question 2: Write an SQL query to find the top 5 customers who placed the highest number of orders.
select c.customer_id, count(o.order_id) as total_orders
from customers c
inner join orders o on c.customer_id=o.customer_id
group by c.customer_id
order by total_orders DESC
LIMIT 5;
```

The query is executed, and the results are displayed in the Data Output tab. The results show the top 5 customers by the number of orders they placed:

customer_id	total_orders
1	31
2	30
3	28
4	19
5	19

The status bar at the bottom indicates "Total rows: 5" and "Query complete 00:00:02.520".

Question 3 (20 marks): Write an SQL query to display a list of orders and the customers who made them. Sort by order date (newest first).

Expected Output:

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following query:

```
-- Question 3: Write an SQL query to display a list of orders and the customers who made them. Sort by order date (newest first)
select o.order_id, o.order_date, c.company_name
from orders o
join customers c on o.customer_id = c.customer_id
order by o.order_date DESC ;
```

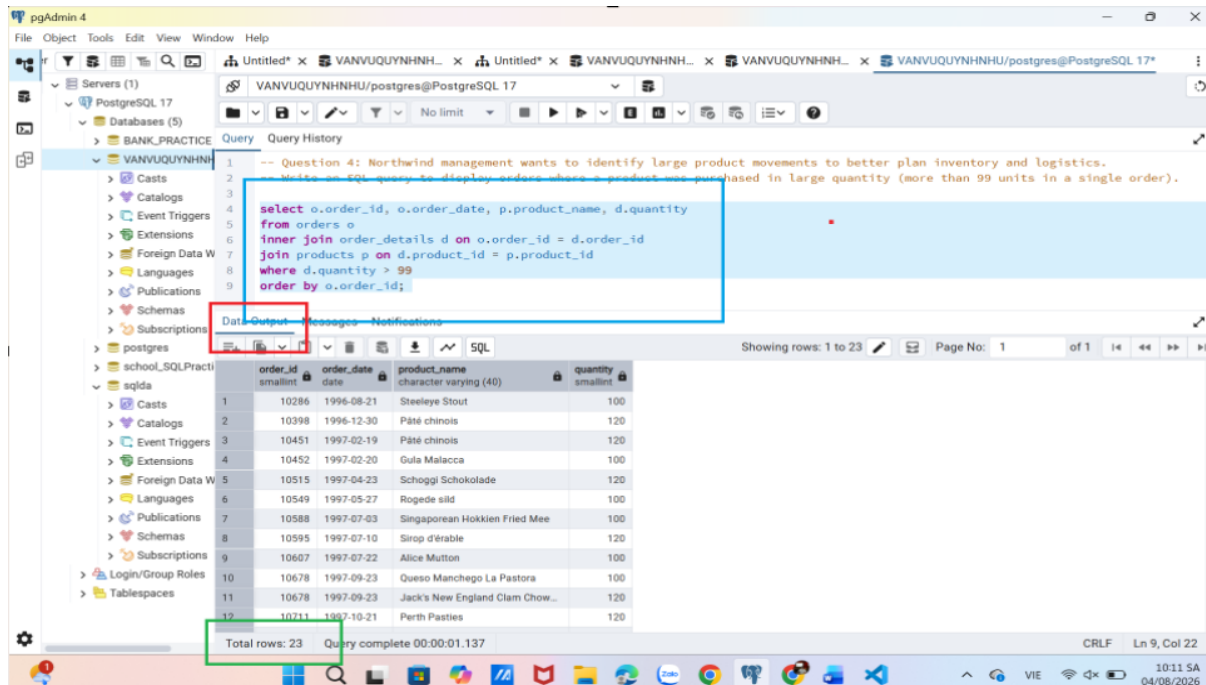
The query is executed, and the results are displayed in the Data Output tab. The results show a list of orders and the customers who made them, sorted by order date (newest first):

order_id	order_date	company_name
1	11074	1998-05-06
2	11077	1998-05-06
3	11076	1998-05-06
4	11075	1998-05-06
5	11071	1998-05-05
6	11073	1998-05-05
7	11072	1998-05-05
8	11070	1998-05-05
9	11068	1998-05-04
10	11069	1998-05-04
11	11067	1998-05-04
12	11065	1998-05-01
13	11066	1998-05-01
14	11064	1998-05-01
15	11063	1998-04-30

The status bar at the bottom indicates "Total rows: 830" and "Query complete 00:00:01.852".

Question 4 (20 marks): Northwind management wants to identify large product movements to better plan inventory and logistics. Write an SQL query to display orders where a product was purchased in large quantity (more than 99 units in a single order).

Expected Output:



The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following query:

```
-- Question 4: Northwind management wants to identify large product movements to better plan inventory and logistics.
-- Write an SQL query to display orders where a product was purchased in large quantity (more than 99 units in a single order).

select o.order_id, o.order_date, p.product_name, d.quantity
from orders o
inner join order_details d on o.order_id = d.order_id
join products p on d.product_id = p.product_id
where d.quantity > 99
order by o.order_id;
```

The query results are displayed in a table with the following columns: order_id, order_date, product_name, and quantity. The results show 12 rows of data.

order_id	order_date	product_name	quantity
1	1996-08-21	Steeleye Stout	100
2	1996-12-30	Pâté chinois	120
3	1997-02-19	Pâté chinois	120
4	1997-02-20	Gula Malacca	100
5	1997-04-23	Schoggi Schokolade	120
6	1997-05-27	Roped sild	100
7	1997-07-03	Singaporean Hokkien Fried Mee	100
8	1997-07-10	Siroop d'érable	120
9	1997-07-22	Alice Mutton	100
10	1997-09-23	Queso Manchego La Pastora	100
11	1997-09-23	Jack's New England Clam Chow...	120
12	1997-10-21	Perth Pasties	120

The status bar at the bottom indicates "Total rows: 23" and "Query complete 00:00:01.137".

Question 5 (20 marks): Northwind management wants to understand which product categories generate the most sales to guide marketing and purchasing decisions. Write an SQL query to calculate the total quantity of products sold for each category.

Expected Output:

pgAdmin 4

Query

```

1  Write an SQL query to calculate the average delivery time (in days) for each shipper.
2  Delivery time = shipped_date - order_date
3  */
4
5  select s.company_name,
6         avg(o.shipped_date - o.order_date) as avg_delivery_days
7  from orders o
8  inner join shippers s on s.shipper_id = o.ship_via
9  group by s.company_name
10 order by avg_delivery_days DESC;
11

```

Data Output

	company_name character varying (40)	avg_delivery_days numeric
1	United Package	9.2349206349206349
2	Speedy Express	8.5714285714285714
3	Federal Shipping	7.4738955823293173

Showing rows: 1 to 3 Page No: 1 of 1

Total rows: 3 Query complete 00:00:01.780

Question 6 (20 marks): Northwind wants to identify the most active customers (customers who place orders most frequently) to target retention campaigns. Write an SQL query to rank customers based on their total number of orders (highest = rank 1). Customers with the same number of orders must have the same rank.

Expected Output:

pgAdmin 4

Query

```

1  /* Question 6: Northwind wants to identify the most active customers (customers who place orders most frequently)
2  to target retention campaigns. Write an SQL query to rank customers based on their total number of orders (highest = rank 1).
3  Customers with the same number of orders must have the same rank.
4  */
5
6  select c.customer_id,
7         c.company_name,
8         count(o.order_id) as total_orders,
9         dense_rank() over(
10             order by count(o.order_id)
11             DESC) as ranking
12  from customers c
13  inner join orders o on c.customer_id = o.customer_id
14  group by c.customer_id, c.company_name
15  order by ranking ASC;
16

```

Data Output

	customer_id character varying (5)	company_name character varying (40)	total_orders bigint	ranking bigint
1	SAVEA	Save-a-lot Markets	31	1
2	ERNSH	Ernst Handel	30	2
3	QUICK	QUICK-Stop	28	3
4	HUNGO	Hungry Owl All-Night Grocers	19	4
5	FOLKO	Folk och fs HB	19	4
6	HILAA	HILARION-Abastos	18	5
7	RATTG	Rattlesnake Canyon Grocery	18	5
8	BERGS	Berglunds snabbköp	18	5
9	BONAP	Bon app'	17	6

Showing rows: 1 to 89 Page No: 1 of 1

Total rows: 89 Query complete 00:00:01.343

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